# Introduction to Relational Algebra

*Basic Relational Algebra Operations*

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## 1. Introduction

Relational Algebra is a **procedural query language** that describes how to retrieve data from a relational database using a sequence of operations. It forms the **mathematical foundation of SQL**. Understanding relational algebra helps students think clearly about how SQL queries work internally.

In this handout, we study the basic operations:

- Selection ($\sigma$)
- Projection ($\pi$)
- Renaming ($\rho$)
- Union ($\cup$)
- Set Difference ($-$)
- Cartesian Product ($\times$)
- Join (Equi-Join, Natural Join)

We also include detailed worked-out examples.

Throughout the handout, we use two sample relations:

$$\text{Student}(\text{SID}, \text{Name}, \text{Dept}, \text{CGPA})$$

$$\text{Enroll}(\text{SID}, \text{Course}, \text{Semester})$$

Example tables:

| SID | Name | Dept | CGPA |
|---|---|---|---|
| 1 | Alice | CSE | 3.9 |
| 2 | Bob | EEE | 3.4 |
| 3 | Carol | CSE | 3.1 |
| 4 | Dave | BBA | 3.6 |

| SID | Course | Semester |
|---|---|---|
| 1 | CSE220 | Spring |
| 1 | CSE230 | Fall |
| 2 | EEE200 | Summer |
| 3 | CSE220 | Fall |

## 2. Selection ($\sigma$)

Selection filters rows based on a condition.

$$\sigma_{\text{Condition}}(\text{Relation})$$

**Example 1.** Find all CSE students.

$$\sigma_{\text{Dept}='CSE'}(\text{Student})$$

**Result:**

| SID | Name | Dept | CGPA |
|---|---|---|---|
| 1 | Alice | CSE | 3.9 |
| 3 | Carol | CSE | 3.1 |

### Multiple conditions

We can combine conditions using logical operators:

$$\sigma_{\text{Dept}='CSE' \ \wedge \ \text{CGPA}>3.5}(\text{Student})$$

**Result:**

| SID | Name | Dept | CGPA |
|---|---|---|---|
| 1 | Alice | CSE | 3.9 |

## 3. Projection ($\pi$)

Projection selects **columns** (attributes).

$$\pi_{\text{Attr1, Attr2}}(\text{Relation})$$

**Example 2.** Extract only the names of all students.

$$\pi_{\text{Name}}(\text{Student})$$

**Result:**

| Name |
|---|
| Alice |
| Bob |
| Carol |
| Dave |

Projection automatically removes duplicates.

## 4. Renaming ($\rho$)

Renaming is used to change the attribute or relation name.

$$\rho_{\text{NewName}}(\text{Relation})$$

or

$$\rho_{\text{NewAttr/OldAttr}}(\text{Relation})$$

**Example 3.** Rename CGPA to Grade.

$$\rho_{\text{Grade/CGPA}}(\text{Student})$$

This is useful when joining relations with same attribute names.

## 5. Union ($\cup$)

Union combines rows of two relations. Both relations must be **union compatible** (same attributes, same domains).

$$R \cup S$$

**Example 4.** Suppose we have two relations:

$$CSEMajors(\text{SID}), \quad EEEminors(\text{SID})$$

To list all SIDs from either group:

$$CSEMajors \cup EEEminors$$

## 6. Set Difference ($-$)

Returns rows present in one relation but not in another.

$$R - S$$

**Example 5.** Students who are enrolled in CSE courses but not in EEE courses.

$$\text{EnrollCSE} - \text{EnrollEEE}$$

## 7. Cartesian Product ($\times$)

Combines every tuple of one relation with every tuple of another.

$$R \times S$$

Not used directly often, but fundamental for joins.

Size becomes:

$$|R \times S| = |R| \cdot |S|$$

**Example 6.** Student $\times$ Enroll produces all possible pairings (not shown due to large size).

## 8. Joins

Joins combine data from two tables based on matching attributes.

Natural Join automatically joins on attributes with the same name.

$$R \bowtie S$$

**Example 7.** Find names of students enrolled in CSE220.

$$\pi_{\text{Name}}(\text{Student} \bowtie \sigma_{\text{Course}='CSE220'}(\text{Enroll}))$$

**Step-by-step solution:**

1. Select students enrolled in CSE220:

$$\sigma_{\text{Course}='CSE220'}(\text{Enroll})$$

Gives:

| SID | Course | Semester |
|---|---|---|
| 1 | CSE220 | Spring |
| 3 | CSE220 | Fall |

2. Natural join with Student on SID:

| SID | Name | Dept | CGPA | Course |
|---|---|---|---|---|
| 1 | Alice | CSE | 3.9 | CSE220 |
| 3 | Carol | CSE | 3.1 | CSE220 |

3. Project only names:

$$\pi_{\text{Name}}(\dots)$$

Final Output:

| Name |
| --- |
| Alice |
| Carol |

## 9. Summary

- Selection ($\sigma$) filters rows.
- Projection ($\pi$) selects columns.
- Renaming ($\rho$) is used for clarity and join conflicts.
- Union and Difference combine or subtract relations.
- Cartesian Product is a fundamental building block of joins.
- Joins combine tuples based on matching attributes.

Relational Algebra forms the core of SQL query planning. Mastering these operations gives deeper insight into how database engines work.